

Jake Callcut

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TECHNICAL SKILLS

- **Languages & Technologies:** Python, SQL, Jupyter, Java
- **Tools & Frameworks:** Pandas, scikit-learn, PyTorch, Transformers, Diffusers, matplotlib, Git, Docker

RELEVANT EXPERIENCE

STMicroelectronics

05/2026 – Present

AI & Machine Learning Intern

- Independently researching and implementing state-of-the-art generative AI methods for synthetic data augmentation within the Data and Knowledge Engineering Team, improving defect detection accuracy on semiconductor wafers by 8%.
- Applying diffusion models with LoRA fine-tuning on proprietary datasets to generate high-fidelity synthetic wafer imagery, increasing available training data by 20%.
- Utilising PyTorch alongside CNN and transformer architectures to evaluate and benchmark data augmentation strategies.

Squiracle

06/2025 – 03/2026

Software Engineer

- Independently designed and developed the company's consumer mobile app and companion business platform, delivering end-to-end functionality from UI/UX to deployment, contributing to investor pitches and funding applications leading to a pre-seed grant with an accelerator.

GI Healthcare Industries

08/2024 – 11/2025

Software Engineer

- Architected and implemented a multi-machine control system with a custom server-client design using Flutter, TypeScript, and Socket.IO, ensuring high fault tolerance and taking the company's new product from TRL 3 to TRL 6, supporting a £250,000 pre-seed round.
- Attended stakeholder product demonstrations, engaging with end-users to gather requirements, observe operational needs, and provide feedback to improve system reliability and usability.

EDUCATION

Heriot-Watt University

09/2022 – 06/2027

MEng Software Engineering — On Track for First Class Honours

- Thrice awarded the Deputy Principal's Award for academic excellence.
- **Relevant Modules:** Data Mining & Machine Learning, Artificial Intelligence & Intelligent Agents, Rigorous Methods for Software Engineering, Big Data Management, Data Structures & Algorithms.

PROJECTS & EXTRA-CURRICULAR

Social Sentiment Analysis for Sector Level Stock Market Prediction | *Dissertation Project (First Class 90/100)*

- Designed and implemented a data pipeline to extract finance data, analyse social media posts for sentiment data using finBERT, and apply machine learning models to predict price movements across various market sectors.
- Evaluated model performance by analysing correlation and classification metrics to assess the predictive power of aggregated social sentiment.

TensoRoom | *Collaborative AI Project*

- Collaborated on building an interior design tool that lets users redesign rooms through natural language, combining diffusion models, segmentation, and VLMs to interpret text prompts and generate photorealistic edits to real room photos.

Judging a Song by Its Cover | *Machine Learning Project*

- Implemented and compared multiple machine learning models and neural networks (kNN, Naive Bayes, MLP, and CNN) to predict music genre based on cover art and metadata.
- Developed algorithmic fine-tuning to optimise models for higher precision and F-Measure without over-fitting.

Lumina | *Personal Project*

- Engineered a smart home hub using Raspberry Pi and WebSockets, enabling reliable real-time communication between IoT devices and a central control system with an intuitive Flutter interface.

ADDITIONAL INFORMATION

- **Certifications:** Bloomberg Market Concepts, MathWorks Deep Learning, GitHub Foundations, DataCamp Associate AI Engineer.
- **Experience:** Freelance Mathematics Tutor & Heriot-Watt Java Lab Assistant.